PROGRAM OFFICE TECHNICAL EVALUATION REPORT Modernization Equipment Acquisition — Competitive Source Selection Report Number: TER-2023-004 Date: September 14, 2023 Classification: UNCLASSIFIED // FOR OFFICIAL USE ONLY

1. PURPOSE

This Technical Evaluation Report (TER) documents the findings of the Source Selection Evaluation Board (SSEB) for the Modernization Equipment Acquisition program. The evaluation assessed proposals submitted in response to Request for Proposal MDA-2023-RFP-017, covering four subsystems: Propulsion, Navigation, Communications, and Power Distribution. The purpose of this report is to provide the Source Selection Authority (SSA) with a comparative technical assessment of all offerors to inform contract award decisions.

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2. EVALUATION METHODOLOGY

Proposals were evaluated against three technical factors: Technical Approach (TA), Management Approach (MA), and Past Performance (PP). Each factor was scored on a scale of 0 to 100. Adjectival ratings were assigned as follows: Outstanding (90–100), Good (75–89), Acceptable (60–74), Marginal (40–59), and Unacceptable (0–39). Scores across factors were weighted equally for the purposes of this evaluation. All evaluations were conducted independently by at least two board members, with adjudicated scores presented here.

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3. OFFERORS EVALUATED

Proposals were received from the following vendors:

Propulsion Subsystem: Apex Defense Systems (Proposal Reference: ADS-PROP-2023) and Vortex Propulsion Group (Proposal Reference: VPG-PROP-2023).

Navigation Subsystem: Meridian Technologies (Proposal Reference: MT-NAV-2023) and Halloran Aerospace (Proposal Reference: HA-NAV-2023).

Communications Subsystem: Consolidated Systems Inc. (Proposal Reference: CSI-COMM-2023) and Halloran Aerospace (Proposal Reference: HA-COMM-2023).

Power Distribution Subsystem: Apex Defense Systems (Proposal Reference: ADS-PWR-2023). One proposal was received. No competitive evaluation was conducted for this subsystem; the single offer was evaluated for technical acceptability only.

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4. PROPULSION SUBSYSTEM EVALUATION

4.1 Apex Defense Systems

Apex Defense Systems submitted a technically detailed proposal centered on their legacy turbine integration platform. The Technical Approach received a score of 82 (Good). Evaluators noted that the proposed turbine seal assembly draws on a mature design with a strong heritage in analogous programs, reducing developmental risk. However, the proposal acknowledged reliance on a single-source supplier for turbine seal components, which evaluators flagged as a supply chain vulnerability. The Management Approach received a score of 78 (Good). The integrated master schedule was realistic and included appropriate schedule margin at key milestones. The Past Performance score was 85 (Good), reflecting successful delivery on two comparable propulsion integration contracts within the past five years, with no significant cost or schedule overruns reported.

Composite Score: 81.7 (Good)

4.2 Vortex Propulsion Group

Vortex Propulsion Group proposed a novel modular turbine architecture intended to reduce long-term maintenance burden. The Technical Approach received a score of 61 (Acceptable). While the concept was innovative, evaluators assessed the maturity level as insufficient for a program with the current schedule constraints. Key enabling technologies were at a Technology Readiness Level (TRL) of 4–5, compared to the program's required TRL of 6 at contract award. The Management Approach received a score of 58 (Marginal). The proposed schedule did not include adequate margin for TRL advancement activities, and the staffing plan identified several critical roles as "to be hired," which evaluators considered a significant execution risk. The Past Performance score was 64 (Acceptable), based on one analogous contract that was completed on time but with a 12% cost overrun.

Composite Score: 61.0 (Acceptable)

Propulsion Recommendation: Apex Defense Systems represents the technically superior offer and is recommended for award.

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5. NAVIGATION SUBSYSTEM EVALUATION

5.1 Meridian Technologies

Meridian Technologies proposed a software-defined navigation architecture leveraging their commercially developed GPS/INS fusion platform, previously adapted for defense applications. The Technical Approach received a score of 88 (Good). Evaluators found the proposed architecture well-suited to the program's interoperability requirements and noted that the software-defined approach allows firmware updates without hardware modification, a significant lifecycle advantage. One evaluator noted a minor concern regarding the proposed verification and validation approach, which relied heavily on simulation rather than hardware-in-the-loop testing at early program phases. This was assessed as low risk given the platform's maturity. The Management Approach scored 84 (Good). The proposal included a detailed risk register with credible mitigation strategies. The Past Performance score was 91 (Outstanding), reflecting three recently completed navigation integration programs with no significant findings.

Composite Score: 87.7 (Good)

5.2 Halloran Aerospace

Halloran Aerospace's navigation proposal was based on a hardware-centric inertial navigation unit with limited software configurability. The Technical Approach received a score of 70 (Acceptable). The design is proven but does not meet the program's interoperability specification without a hardware modification that Halloran estimated would add four months to the schedule. Evaluators assessed this as a moderate technical risk. The Management Approach scored 72 (Acceptable). The schedule was achievable but provided no margin for the hardware modification identified above. Past Performance scored 75 (Good), based on two completed programs with satisfactory delivery and minor cost growth on one contract.

Composite Score: 72.3 (Acceptable)

Navigation Recommendation: Meridian Technologies represents the technically superior offer and is recommended for award.

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6. COMMUNICATIONS SUBSYSTEM EVALUATION

6.1 Consolidated Systems Inc.

Consolidated Systems Inc. proposed a waveform-agnostic communications suite built around their CSI-7 radio platform, which has been fielded on several analogous programs. The Technical Approach received a score of 91 (Outstanding). Evaluators assessed the proposed design as fully compliant with all interface control document requirements and noted that the CSI-7 platform's waveform agnosticism provides significant flexibility for future capability upgrades without re-procurement. The Management Approach scored 86 (Good). The proposal demonstrated strong systems engineering discipline, with a well-structured CDR entry criteria checklist and a realistic risk retirement schedule. Past Performance scored 89 (Good), reflecting four contracts completed within cost and schedule, with one program receiving a Contractor Performance Assessment Report (CPAR) rating of Exceptional.

Composite Score: 88.7 (Good)

6.2 Halloran Aerospace

Halloran Aerospace submitted a communications proposal based on a fixed-waveform radio design. The Technical Approach scored 66 (Acceptable). The design meets current interface requirements but lacks the waveform flexibility identified as a key evaluation discriminator. Evaluators noted that future waveform upgrades would require hardware replacement rather than software updates, increasing long-term lifecycle cost. The Management Approach scored 69 (Acceptable). The proposal included a schedule that evaluators found optimistic at the critical design and fabrication phases. Past Performance scored 74 (Acceptable), based on two completed programs, one of which experienced a six-month schedule slip attributed to subcontractor performance issues.

Composite Score: 69.7 (Acceptable)

Communications Recommendation: Consolidated Systems Inc. represents the technically superior offer and is recommended for award.

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7. POWER DISTRIBUTION SUBSYSTEM EVALUATION

One proposal was received for the Power Distribution subsystem from Apex Defense Systems. The proposal was evaluated for technical acceptability only; no competitive comparison was conducted. The Technical Approach scored 74 (Acceptable). The proposed design is straightforward and leverages commercial off-the-shelf components for the primary distribution bus. Evaluators flagged the capacitor bank as relying on a single qualified supplier, representing a supply chain risk that should be tracked as a program-level risk item. The Management Approach scored 78 (Good). Past Performance scored 80 (Good).

Composite Score: 77.3 (Good)

Power Distribution Recommendation: Apex Defense Systems' proposal is technically acceptable and recommended for award. The single-source capacitor bank risk shall be tracked as a risk item and a mitigation plan shall be required within 60 days of contract award.

8. SUMMARY OF SCORES

Subsystem	Vendor	Technical Approach	Management Approach	Past Performance	Composite
Propulsion	Apex Defense	82 (Good)	78 (Good)	85 (Good)	81.7 (Good)
Propulsion	Vortex Propulsion Group	61 (Acceptable)	58 (Marginal)	64 (Acceptable)	61 (Acceptable)
Navigation	Meridian Technologies	88 (Good)	84 (Good)	91 (Outstanding)	87.7 (Good)
Navigation	Halloran Aerospace	70 (Acceptable)	72 (Acceptable)	75 (Good)	72.3 (Acceptable)
Communications	Consolidated Systems Inc.	91 (Outstanding)	86 (Good)	89 (Good)	88.7 (Good)
Communications	Halloran Aerospace	66 (Acceptable)	69 (Acceptable)	74 (Acceptable)	69.7 (Acceptable)
Power Distribution	Apex Defense Systems	74 (Acceptable)	78 (Good)	80 (Good)	77.3 (Good)

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9. CONCLUSIONS AND RECOMMENDATIONS

The SSEB recommends award to Apex Defense Systems for the Propulsion and Power Distribution subsystems, Meridian Technologies for the Navigation subsystem, and Consolidated Systems Inc. for the Communications subsystem. These recommendations are based solely on technical merit as documented herein. Price evaluation was conducted separately and is documented in the Price Evaluation Report (PER-2023-004).

The SSEB notes that the single-source supply chain risk identified in both Apex Defense Systems proposals (turbine seals for Propulsion and capacitor bank for Power Distribution) should be elevated to program-level risk tracking regardless of award outcome. The risk retirement approach proposed by Apex for the Power Distribution subsystem is credible; the Propulsion subsystem proposal did not include a comparable mitigation.

Prepared by: Source Selection Evaluation Board, Modernization Equipment Acquisition Program Office Approved by: [SSA Signature Block]

A few things deliberately built in for eval questions:

- Vortex Propulsion Group and Halloran Aerospace were evaluated and lost — "was Vortex evaluated?" is answerable; "was [made-up vendor] evaluated?" is a trap

- Apex appears on two subsystems with different risk profiles — cross-doc synthesis with the CSV
- The capacitor bank single-source risk appears in both this doc and the CSV — intentional thread to pull on
- No Targeting or Fuel Management subsystem anywhere — clean trap question gap